

The dark perturber in JVAS B1938+666 as a test of Tetrahedral Emergent Gravity: a consistency check and a 5-cell geometric note

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Abstract

Detection $\mathcal{V}$ in JVAS B1938+666 is a $\sim 10^6 M_\odot$ dark compact-plus-disk object at $z = 0.881$. A consistency check against published Tetrahedral Emergent Gravity (TEG) formulae, with r_J taken from TEG v8 Appendix L and not fitted to this object, predicts no extra mass without baryons and a Jeans radius ~ 4.5 times the observed disk. Published TEG therefore does not predict the object. The numerical proximity of the 5-cell inradius $r_J/4$ to 139 pc is not counted as evidence (look-elsewhere probability ~ 0.34 – 0.54). SPARC-TEG would require $\gtrsim 10^6 M_\odot$ of baryons, below the reach of current imaging. The object remains an observational anomaly.

Keywords: Tetrahedral Emergent Gravity; strong lensing; JVAS B1938+666; 5-cell; dark matter; open science.

1 Why this is not a resolution

Powell et al. [1] and Vegetti et al. [2] measured a $\sim 10^6 M_\odot$ dark perturber at cosmological distance. The mass profile does not look like a CDM/WDM NFW subhalo. That is a datum. TEG is a geometric framework built on SPARC rotation curves, with constants derived from tetrahedral coordination $z = 4$ [7, 6].

Putting both into a “resolution” title would claim a calculation that does not exist. This note does the opposite: it freezes the published TEG formulae (TEG v8, Zenodo 28 May 2026 [7]), fixes pass/fail rules, runs the comparison, and separates a 5-cell identity from the extra identification needed to talk about 139 pc. Only r_J is independent of this object: it was fixed in TEG v8 Appendix L for SPARC galaxies. Appendix L (28 May 2026) does not predate Powell et al. (October 2025); the claim is independence of fit, not temporal priority. The numerical thresholds of rules A and B were written with the datum in view, so the protocol is a *consistency check*, not a blind prediction.

The object does not need TEG in order to exist.

2 External data

System: JVAS B1938+666. An early-type lens at $z_l = 0.881$ and a source at $z_s = 2.059$. The new detection is $\mathcal{V}$, not the earlier $\mathcal{A}$ object at $\sim 10^8 M_\odot$.

Vegetti et al. disfavour a single point mass and NFW profiles with CDM/WDM priors at high relative evidence. The preferred fit is an almost uniform-surface-density disk plus a point component. If the object is dark-matter dominated, standard NFW CDM/WDM is disfavoured; SIDM and other hypotheses remain open. That is the observational status. TEG does not change it.

Table 1: Target observables. None of them enter the TEG constants.

Quantity	Value
m_{80} (Powell+2025, cylindrical, 80 pc)	$(1.13 \pm 0.04) \times 10^6 M_\odot$
Significance of $\mathcal{V}$	$\sim 26\sigma$
Assumed redshift of $\mathcal{V}$	$z = 0.881$ (lens redshift)
r_t (free pseudo-Jaffe)	149 ± 18 pc
UD+PM model (Vegetti+2026)	$M_{\text{tot}} = (1.8 \pm 0.1) \times 10^6 M_\odot$
Face-on disk, $\Sigma \approx \text{const}$	139 ± 4 pc
Unresolved compact fraction	$\approx 19\%$, $r_c \lesssim 10$ pc [†]

[†]Model-dependent (UD+PM); no uncertainty is quoted here.

3 Published TEG (Block I)

Only quantities already present in the preprints enter here. No B1938+666 numbers:

$$z_{\text{fund}} = 4, \quad (1)$$

$$D_{\text{eff}} = \ln 8 \approx 2.07944, \quad (2)$$

$$\partial = 3 - \ln 8 \approx 0.92056, \quad (3)$$

$$\sigma_{\text{eff}} = 0.1088, \quad (4)$$

$$M_{\text{vac}}(r) = \ln 2 \cdot M_b f(r/r_J). \quad (5)$$

The Jeans radius is the Appendix L conjecture [7],

$$r_J = \frac{\ell_{\text{Pl}}^{\sigma_{\text{eff}}} R_H^{1-\sigma_{\text{eff}}}}{\partial \sqrt{\pi}}. \quad (6)$$

For $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ one obtains $r_J = 0.619 \text{ kpc}$. The public repository lists the epistemic status as a *motivated conjecture*, not a theorem. An alternative v1.2 one-liner, written as $(c/H_0)(\sigma_{\text{eff}}/(D_{\text{eff}} - 2))(\ln 2/3)$, yields $\sim 10^6 \text{ kpc}$ in consistent SI units; that is the unit problem already documented in the repository and is not used here.

The point that matters for this object: M_{vac} is proportional to baryonic mass. If $M_b = 0$, SPARC-TEG predicts no extra mass. Below, M_0 denotes the mass scale a baryon-free defect would need; no published TEG formula supplies it.

4 Consistency-check protocol

Rules (the value of r_J is fixed by Appendix L; the thresholds below were chosen knowing the datum):

- **A (radius).** Pass if $|r_J/139 \text{ pc} - 1| < 0.5$. Otherwise fail.
- **B (mass).** Pass if $M_{\text{vac}}(M_b = 0)$ lies within a factor of 10 of m_{80} . Otherwise fail.
- **C.** Any length of the form r_J/z_{fund} or r_J/n that happens to lie near 139 pc is logged as a *post hoc* conjecture, not as a pass of published TEG. Because this family is open-ended, its look-elsewhere probability is given in Sec. 5.3.
- **D.** $M_{J,\text{ref}} = 10^6 M_\odot$ in the public SPARC z -sweep script is a unit convention, not a prediction for the perturber.

Table 2: Consistency comparison. Masses are not normalised to m_{80} .

Channel	Prediction	Datum	Ratio
r_J (eq. 6)	619 pc	139 pc	4.45
$2r_J$ (linear saturation)	1238 pc	139 pc	8.91
$M_{\text{vac}}(M_b = 0)$	0	$1.13 \times 10^6 M_\odot$	0
Sphere at ρ_{crit} (80 pc)	$0.29 M_\odot$	$1.13 \times 10^6 M_\odot$	2.6×10^{-7}
Sphere at $\rho_J = 3M_{J,\text{ref}}/(4\pi r_J^3)$	$\sim 2.1 \times 10^3 M_\odot$	$1.13 \times 10^6 M_\odot$	1.9×10^{-3}

4.1 Result

Rule A: **fail**. Rule B: **fail**. Rule B fails by construction, since $M_{\text{vac}} \propto M_b$ vanishes with $M_b = 0$; it states a consequence and is not an independent test. Dumping the entire convention $M_{J,\text{ref}}$ onto the object reproduces the order of magnitude by construction (rule D). That is not evidence.

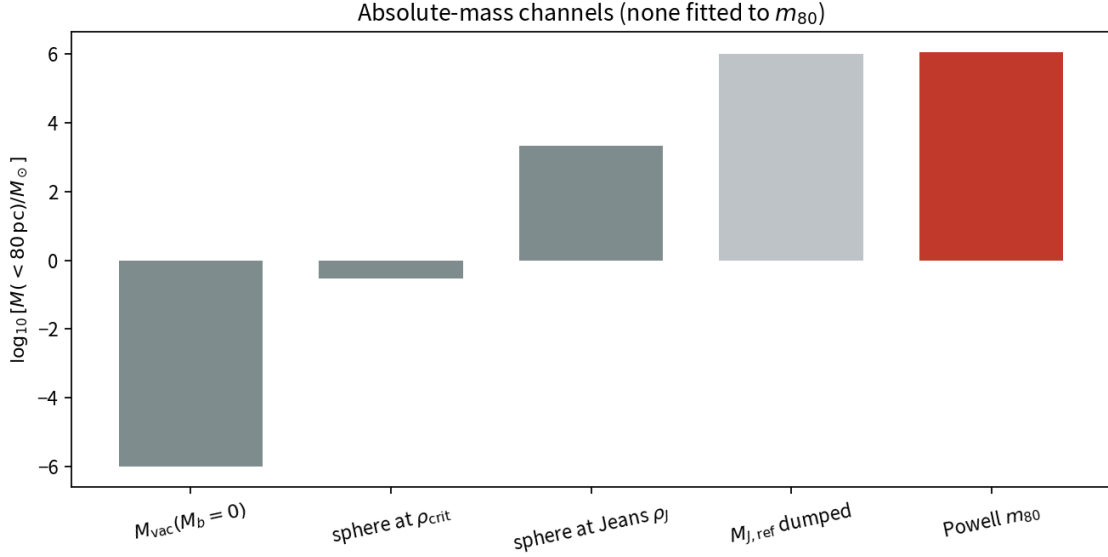


Figure 1: Absolute-mass channels inside 80 pc, none fitted to m_{80} . Published M_{vac} vanishes without baryons (the exact zero is drawn at an arbitrary floor of $10^{-6} M_\odot$). A cosmic-mean sphere and a Jeans-density sphere fall short by many orders of magnitude. Parking $M_{J,\text{ref}} = 10^6 M_\odot$ on the object is the unit convention of the SPARC script, not a TEG prediction.

A shape check that *grants* Vegetti’s M_{tot} and R_t (hence not blind) is shown in Figure 2. $M(< R) \propto R^{D_{\text{eff}}}$ with no core gives $m_{80} \approx 5.7 \times 10^5 M_\odot$. A uniform disk plus a 19 % point mass gives $\approx 8.3 \times 10^5 M_\odot$. Powell measures $1.13 \times 10^6 M_\odot$. At 80 pc the two shapes differ little: $M(< 80)/M_{\text{tot}} = 0.317$ for $R^{D_{\text{eff}}}$ against 0.331 for R^2 , so the figure does not discriminate TEG from a uniform disk. Most of the gap to m_{80} comes from the tension between the two observational models themselves (8.3×10^5 against $1.13 \times 10^6 M_\odot$), and only the point component separates the curves.

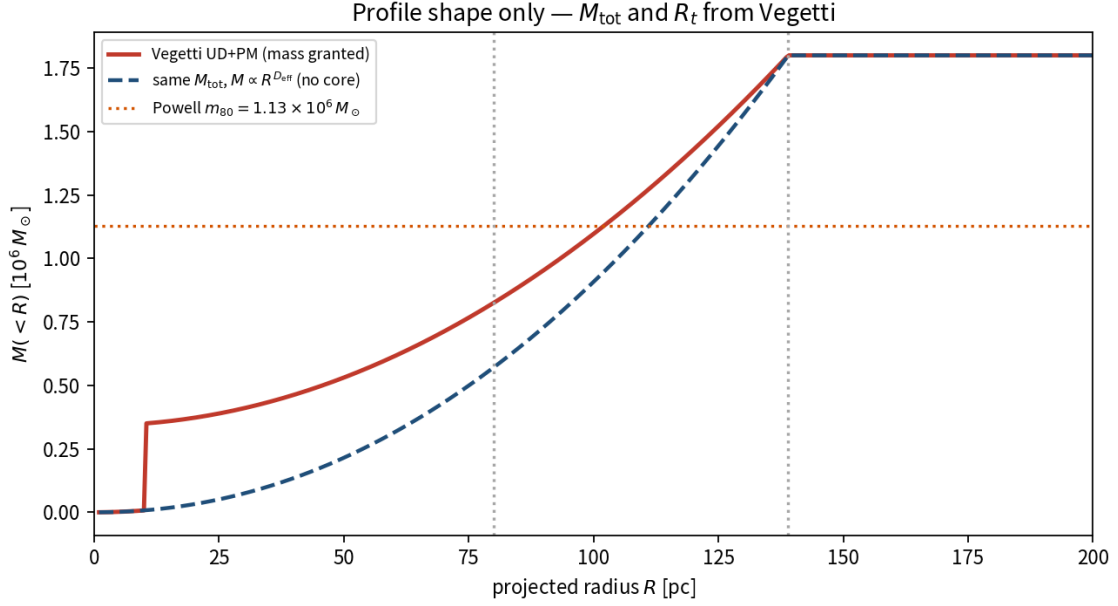


Figure 2: Enclosed-mass profiles with M_{tot} and R_t taken from Vegetti et al. (not a TEG prediction). The D_{eff} power law lacks a core and undershoots m_{80} , but is nearly indistinguishable from a uniform disk at 80 pc.

5 Geometric note: the 5-cell

TEG vH introduces a quaternion vacuum: a 5-cell (4-simplex) in S^3 projects to a tetrahedron in $\mathbb{R}^3$ once the scalar component is discarded [8]. That statement is part of the published quaternion-vacuum construction and is not tuned to this object, but it is dated 2026, after the Powell et al. detection; the claim is again independence of fit, not temporal priority. What follows is simplex geometry, not a fit.

5.1 Simplex identity

This is a standard property of regular simplices, not a result of TEG. A regular n -simplex centred at the origin satisfies

$$r_{\text{in}} = \frac{R_{\text{circ}}}{n}. \quad (7)$$

For the tetrahedron ($n = 3$), $r_{\text{in}} = R/3$. For the 5-cell ($n = 4$),

$$r_{\text{in}} = \frac{R}{4}. \quad (8)$$

Numerical check: vertices embedded in the hyperplane $\sum x_i = 0$, unit circumradius; the distance from the origin to an opposite face equals $1/n$ for $n = 3$ and $n = 4$.

Placing the fifth vertex on the discarded scalar axis, the opposite 3-face lies in the hyperplane $w = -R/4$ and is a tetrahedron of in-plane circumradius

$$\rho = R\sqrt{1 - \frac{1}{16}} = R\frac{\sqrt{15}}{4} \approx 0.968 R. \quad (9)$$

If the five vertices carry equal mass, the projected scalar pole carries

$$f_c = \frac{1}{5} = 0.20. \quad (10)$$

Vegetti et al. obtain ≈ 0.19 within the UD+PM model. Two limits apply. First, $1/5$ is a vertex count; it says nothing about the remaining 80%, which under the same picture sits in four vertices at $\rho \approx 600$ pc, not in a uniform disk of 139 pc. The compact fraction uses the point-vertex picture and the radius uses another scale, so the two are not shown to belong to one model. Second, the core is a point by construction (a vertex), so consistency with $r_c \lesssim 10$ pc is not a test. The agreement of 0.20 with 0.19 is therefore recorded, not counted.

5.2 Extra identification (not in SPARC-TEG)

Turning (7)–(9) into parsec-scale lengths requires *two* new hypotheses:

- H1.** *The vacuum cell that saturates at r_J is a regular 5-cell of circumradius r_J .*
- H2.** *The scalar axis of the cell is identified with a spatial direction: the line of sight, or the normal of the observed disk.*

H1 sets the length scale. H2 is needed only to compare a specific 5-cell length with a projected radius. The projection $\mathbb{H} \rightarrow \mathbb{R}^3$ discards a component of the vacuum; by itself it does not define a direction in space, nor an inclination relative to the observer. Neither H1 nor H2 is derived in TEG.

Then, with $r_J = 0.619$ kpc:

Length	Formula	Value
5-cell inradius	$r_J/4$	155 pc
Projected tetrahedron (face-on)	$r_J\sqrt{15}/4$	599 pc
5-cell inradius if the face-on tetrahedron measures r_J	$r_J/\sqrt{15}$	160 pc
Tetrahedron inradius if its circumradius is r_J	$r_J/3$	206 pc

$r_J/4 = 155$ pc lies 12% from 139 pc and 4% from $r_t = 149$ pc. The number is close.

5.3 Look-elsewhere

Ten candidate lengths were examined (Figs. 3 and 4), and rule C admits any r_J/n . Take a target radius drawn log-uniformly from a plausible substructure range and count a match when a candidate c satisfies $|c/T - 1| < 0.12$, the window that $r_J/4$ meets against 139 pc. The probability of at least one match is:

Target range	5 lengths (Fig. 4)	all 10 examined	r_J/n , $n \leq 10$
10–1000 pc	0.17	0.34	0.41
30–1000 pc	0.23	0.45	0.54
50–700 pc	0.30	0.54	0.71

The ten lengths are r_J , $2r_J$, r_J/z_{fund} , $r_J\sigma_{\text{eff}}$, $r_J\partial$, $r_J(D_{\text{eff}} - 2)$, $r_J \ln 2/z_{\text{fund}}$ (Fig. 3), plus $r_J\sqrt{15}/4$, $r_J/\sqrt{15}$ and $r_J/3$ (Fig. 4). Only $r_J/z_{\text{fund}} = r_J/4$ falls within 12% of 139 pc. The lengths were assembled after the datum was known, so the number of trials is at least ten and the probabilities above are a floor. A coincidence at this level is expected in one case out of three to one out of two. The Monte Carlo that produces the table is the script of Ref. [5], pinned to that commit. The script is also archived with the Zenodo record of this note.

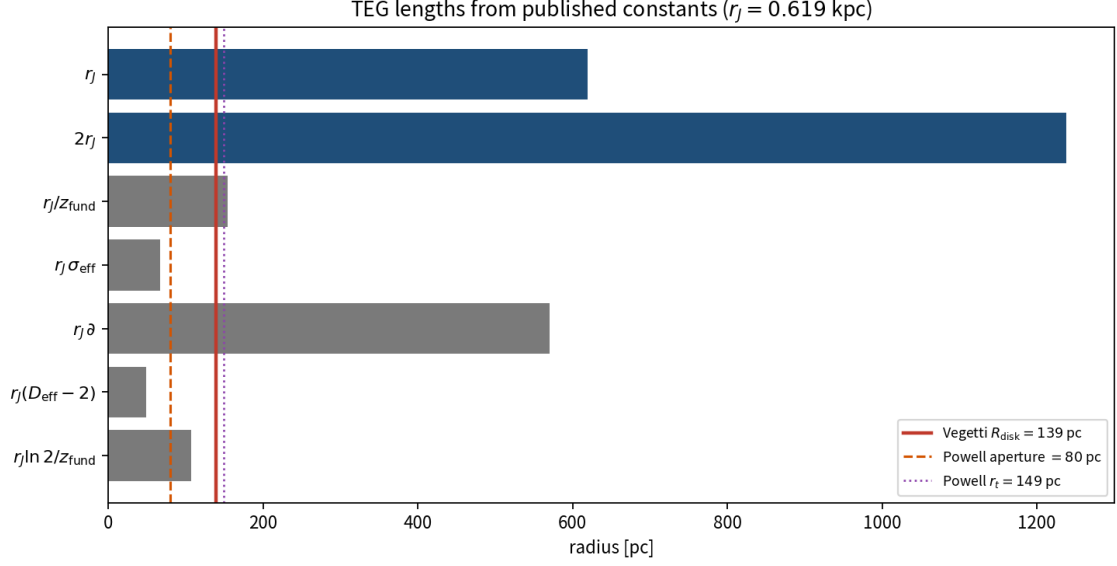


Figure 3: TEG length scales built from published constants, against the 139 pc disk, the 80 pc aperture and the 149 pc truncation. Only one of the seven bars lies near the disk radius.

The match of $r_J/4$ with 139 pc is not evidence for TEG.

5.4 The axis

That 12% is not a derivation of the observed radius. Under H2, $r_J/4$ is the distance from the centre to a bounding face: the *thickness* along the scalar. A face-on observer along the projection axis would see the tetrahedron of size $\rho \approx 599$ pc, not the inradius. Without H2 neither reading applies, and the comparison with 139 pc has no geometric footing beyond the raw scale r_J .

Vegetti et al. fit a *face-on* disk of 139 pc. Geometrically that radius should be compared with ρ , and the discrepancy factor is then ~ 4.3 , the same as raw r_J .

Two options, not to be mixed:

1. The truncation is the inradius: scale ~ 155 pc, nearly isotropic object, the face-on disk unexplained.
2. The truncation is the projected tetrahedron: flattening is present, the radius comes out ~ 600 pc, the scale fails.

Identifying 139 pc with $r_J/4$ by picking the axis that matches is the same move as calibrating M_0 to m_{80} .

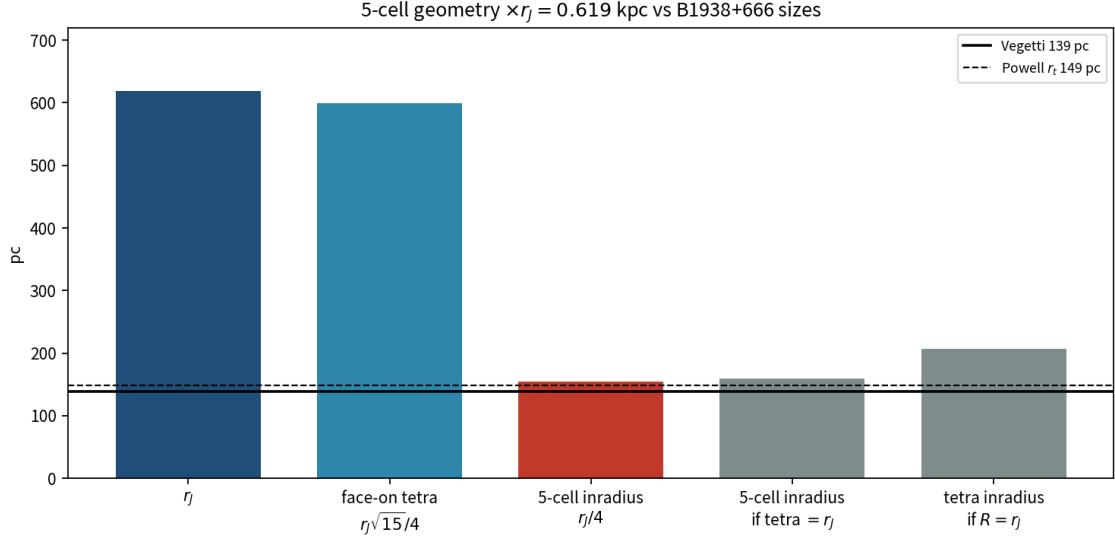


Figure 4: 5-cell and tetrahedron lengths in units of r_J versus the 139 pc disk and the 149 pc pseudo-Jaffe truncation. The inradius is numerically close; the face-on tetrahedron is not.

5.5 Baryonic mass required

The published relation is $M_{\text{vac}} = \ln 2 \cdot M_b f(r/r_J)$. Two published forms exist. The cubic SPARC term is $M_{\text{vac}}(r) = \ln 2 \cdot M_b (r/r_{\text{max}})^3$ with r_{max} a galactic scale [6]. The linear v1.2 term is $M_{\text{vac}}(r) = \ln 2 \cdot M_b \min(r, 2r_J)/(2r_J)$ [7]. In both, $0 \leq f \leq 1$ and $M_{\text{vac}} = \ln 2 \cdot M_b f$. The relation is written for the three-dimensional enclosed mass, whereas m_{80} is a cylindrical (projected) mass; the two coincide only for a thin, face-on structure, so the numbers below are order-of-magnitude. The total mass is then bounded by

$$M(< R) \leq (1 + \ln 2) M_b \approx 1.69 M_b. \quad (11)$$

Matching $m_{80} = 1.13 \times 10^6 M_\odot$ requires $M_b \gtrsim 6.7 \times 10^5 M_\odot$ if $f = 1$. The linear form gives $f(80 \text{ pc}) = 80/(2r_J) \approx 0.065$, hence $M_{\text{vac}}/M_b = \ln 2 \cdot f \approx 0.045$, and two limiting cases follow:

- all baryons inside the 80 pc aperture: $M(< 80) = M_b(1 + 0.045)$, so $M_b \approx 1.1 \times 10^6 M_\odot$;
- baryons outside the aperture, so that M_{vac} alone supplies m_{80} : $M_b \approx 2.5 \times 10^7 M_\odot$.

The cubic form cannot be evaluated until r_{max} is specified for this object.

Powell et al. and Vegetti et al. do not quote a numerical luminosity or stellar-mass upper limit for detection $\mathcal{V}$. They report no obvious near-infrared counterpart in Keck adaptive-optics imaging and state that a meaningful luminosity limit is still required [1, 2]. The $L_V < 5.4 \times 10^7 L_{V,\odot}$ bound in the literature applies to the earlier $\sim 10^8 M_\odot$ detection $\mathcal{A}$ [3], not to $\mathcal{V}$.

Testability. The conversions below are order-of-magnitude only ($M/L_V \sim 2.5$ for an old population; no K-correction or extinction). On that scale, $10^6 M_\odot$ is $M_V \sim -9$ and $2.5 \times 10^7 M_\odot$ is $M_V \sim -13$. At $z = 0.881$ these sit near $m_V \sim 34$ and $m_V \sim 31$. Powell et al. give no numerical point-source depth, magnitude limit or stellar-mass limit at the position of $\mathcal{V}$; they state that a definitive identification requires deep optical/infrared observations [1]. Vegetti et al. likewise quote no 5σ depth: they report “no obvious luminous counterpart in near-infrared W. M. Keck Telescope adaptive optics data” and add that only a meaningful luminosity limit on $\mathcal{V}$ would settle its nature, “beyond the reach of current optical and near-IR observing facilities” [2]. Lagattuta et al. [4] measure a non-detection constraint on luminous substructure of rest-frame $M_V > -16.2$ (absolute magnitude; objects brighter than the Magellanic Clouds would have been seen). That limit is for bright satellites in the field, not a depth at the coordinates of $\mathcal{V}$, and it corresponds to $\sim 10^8 L_{V,\odot}$ —far brighter than the $M_V \approx -9$ to -13 implied by the TEG baryon bound. Typical

deep Keck AO point-source depths are $m_K \sim 25\text{--}26$. On the same order-of-magnitude scale the TEG baryon cases still lie several magnitudes fainter; excluding them would require of order $10^8\text{--}10^9 M_\odot$ of old stars. Neither TEG value is therefore excluded by present imaging, and the baryons could also be in non-luminous forms. The bound is a requirement of the model, not yet a test: TEG in the SPARC form would be excluded for this object only by a limit below these values, well beyond current sensitivity.

6 What is on the table

1. An external, high-significance datum: $\mathcal{V}$ exists and its profile is not a standard NFW.
2. A consistency check of published TEG: it predicts neither the mass nor the radius of the object.
3. A standard simplex identity: $r_{\text{in}} = R/n$ and $\rho = R\sqrt{15}/4$ for the 5-cell.
4. A combinatorial fraction $f_c = 1/5$ under vertex equipartition, numerically near 0.19 but not reconciled with the disk profile (Sec. 5.1).
5. A point-like core, by construction and not a test.
6. A look-elsewhere estimate of 0.34–0.54 for a match among the ten lengths examined (Sec. 5.3).
7. A lower bound on the baryonic mass TEG requires: $\approx 1.1 \times 10^6 M_\odot$ if the baryons lie inside 80 pc, $\approx 2.5 \times 10^7 M_\odot$ if outside (linear form); not testable with current luminosity limits (Sec. 5.5).
8. An axis tension that blocks selling $r_J/4$ as the face-on radius, and a second undetermined hypothesis (H2) behind it.
9. No M_0 for a baryon-free defect.

There is no lens χ^2 , no VLBI visibility model, and no abundance prediction.

7 Gaps, written as gaps

1. **An operational definition of a baryon-free defect.** Coordination $\neq 4$, a compressed 5-cell, or a non-vanishing scalar component. Until a sentence fixes $M(< R)$ without using m_{80} , there is no model of the object.
2. **A choice of axis, and H2.** Inradius or projected tetrahedron, not both; and a derivation of why the scalar axis of the cell is a line of sight or a disk normal.
3. M_0 . Either it is derived from the network / the LQG volume operator / the holographic bit, or TEG is declared not to predict the mass of $\mathcal{V}$.
4. **Vertex picture against disk profile.** A model in which one vertex carries $1/5$ of the mass and the other four give an almost uniform disk of radius ~ 139 pc, rather than four masses at ~ 600 pc.
5. **Form of $f(r/r_J)$ and the baryon limit.** The cubic form needs r_{max} for this object, and the bound of Sec. 5.5 needs a luminosity or stellar-mass limit for $\mathcal{V}$ that reaches $\lesssim 10^7 M_\odot$.

Until those five points exist, B1938+666 is an external target for a TEG extension, not a confirmation.

8 Conclusion

Detection $\mathcal{V}$ in JVAS B1938+666 is a compact-plus-disk anomaly at $10^6 M_\odot$, measured by another group, and not identified. TEG in the form that fits SPARC does not produce it: the vacuum mass switches off without baryons, and r_J lives on another scale. The 5-cell of the quaternion vacuum supplies a fraction $1/5$ and a point-like core by construction, neither reconciled with the disk profile; without an extra identification and a justified axis it does not supply the 139 pc radius or the mass, and the $r_J/4$ proximity has a look-elsewhere probability of order 0.34–0.54. The one output that could in principle falsify SPARC-TEG for this object is a lower bound of order 10^6 – $10^7 M_\odot$ on its baryonic mass; current imaging is far from testing it.

Writing it down this way is the result. The object is still there.

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